

COMP364 Quiz 6

Question 1. Masked self-attention (5 points)

In the phrase *masked self-attention*, briefly describe the meaning of “masked” and “self”.

Solution.

“Masked” means that tokens cannot attend to future tokens—the future is *masked*. “Self” means that a token may attend to itself.

Question 2. Single-head attention operator (5 points)

List the major quantities employed by a single-head attention operator, in the approximate order of computation.

Solution.

token embedding, position embedding, input representation, query vectors, key vectors, value vectors, attention weights, attention-weighted value vectors, projected attention output, post-residual output representation.

Question 3. Multi-head attention operator (5 points)

Describe the difference between single-head and a multi-head attention in one sentence.

Solution.

Multi-head attention splits the input representation by projecting it into H smaller representations, computing standard attention on each, then adding the results to produce the projected attention output.

Question 4. Attention layer (5 points)

List the major quantities employed by an attention layer, in the order of computation.

Solution.

Masked self-attention, layer normalization, feed-forward network, layer normalization.

Question 5. Transformer (5 points)

List the major quantities and components of a transformer neural network architecture, in their order of computation.

Solution.

token embedding, position embedding, input representation, a sequence of attention layers, unembedding, softmax, token logits, token prediction

Question 6. Training targets (5 points)

A certain transformer uses lower case letters from the English alphabet as its only tokens. The following sequence is its only training data:

yymfndifhdjsjsjshfgkgdjdvbkfrbsbivb

The transformer employs a block size of 5, and it is being trained using batches of size 2. One particular batch of training data employs the inputs

$$x = \begin{pmatrix} \text{hdjsj} \\ \text{jdvbk} \end{pmatrix}.$$

For this batch, what value of the targets y should be employed?

Solution.

We need sequences of the training data that match the inputs except shifted right by one position. This gives

$$y = \begin{pmatrix} \text{djsjs} \\ \text{dvbkf} \end{pmatrix}.$$

Question 7. Predictions and loss calculation

A transformer uses the vocabulary $\{\mathbf{a}, \mathbf{b}\}$ with IDs $\{\mathbf{a} : 0, \mathbf{b} : 1\}$. It processes a sequence of length 3, producing the final logits

$$\begin{pmatrix} 3.1 & -0.5 \\ 0.2 & 0.7 \\ -1.1 & -0.1 \end{pmatrix}.$$

(a) **(5 points)** What are the predicted probabilities for the next token, and which token will be output?

Solution.

Prediction uses logits for the final sequence position only, so we compute a softmax of $(-1.1, -0.1)$, yielding

$$\left(\frac{e^{-1.1}}{e^{-1.1} + e^{-0.1}}, \frac{e^{-0.1}}{e^{-1.1} + e^{-0.1}} \right) = (0.27, 0.73).$$

The highest probability corresponds to token ID 1, so the output is **b**.

(b) **(5 points)** Suppose instead that the same logits above were produced by training on a single input sequence $x = \mathbf{aba}$ and target $y = \mathbf{abb}$. What is the contribution of the final token to the cross-entropy loss?

Solution.

The total cross-entropy loss is $-\sum_i \log p_{y_i}$, where y_i is the token ID of the i th target token. The question asks for the contribution from the final token only. The final target is **b** (token ID 1). From the previous question, we know the computed probability of **b** was 0.73. So the contribution of this token to the loss is

$$-\log 0.73 = 0.14.$$

Total number of points: **40**